

Discrete Mathematics I

Sets

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Sets

A **set** can be informally thought of as an ‘unordered collection of objects’. The ‘members’ of a **set** (‘collection’) are called the **elements of** the **set**.

Stages of set theory

- **naive set theory:** Informal, does not use a formalised language. It assumes that any property expressed in natural language can be used for defining sets. This informal, ‘naive’ approach leads to contradictions.
- **axiomatic (modern) set theory:** A formal, axiomatised theory of sets, which aims to get rid of the contradictions in naive set theory. It uses a formal, axiomatic approach:
 - it lays down a set of **axioms** (statements that are accepted automatically as true without any proof);
 - the theory is deduced from the axioms by applying the rules of logic.

Sets: basics

Informally: We can think of a **set** as an **unordered collection** of objects. The **objects in this collection** are called the **elements** of the set.

Formal definition: Zermelo-Fraenkel set theory axioms (see Appendix).

Basic, undefined concepts (predicates) in set theory

- **set**
- $x \in A$: x is an **element of** set A (or x **belongs to** A).

Note: The elements of a set can be any kinds of 'objects'.

Axiom of extensionality

Two sets are equal if and only if they contain exactly the same elements.

Note:

- Elements of a set **do not have 'multiplicities'**: an object is either an element of a set or not, it cannot be an element twice...
- **no order** exists on the elements (by default, unless further structures are imposed on the set)

Defining sets using the roster notation

By the Axiom of extensionality in order to **define a set** we need to **specify what its elements are**.

Defining a (finite) set by listing its elements (roster notation):

A finite set can be defined by listing its elements between a pair of curly brackets $\{ \}$, separated by commas.

Examples:

- $\{a\}$ denotes the set which contains only a single element a ;
- $\{a, b\}$ is the set containing exactly the elements a and b ;
- $\{5, c, \{a, b, c\}, *, \{b\}\}$ is the set containing exactly the elements:

$5, c, \{a, b, c\}, *$ and $\{b\}$.

Note:

- the **order** the elements are listed **does not matter**, e.g. $\{a, b\} = \{b, a\}$;
- elements have **no multiplicity**, e.g. $\{a, b\} = \{a, a, b, b, b\}$.

Some special sets

Definition (empty set)

The set containing no elements is called the **empty set** and denoted by $\emptyset$ or $\{\}$.

Note

- Please note that $\emptyset \neq \{\emptyset\}$!
- By the Axiom of extensionality the empty set is unique.

Definition (system of sets)

A set in which all the elements are also sets is sometimes also called a **system of sets**.

Examples:

- $\{\{2, 3, 5\}, \{4, 8\}, \{1\}, \{1, 3\}\}$
- $\{\{d, e, f\}, \{1, 4\}, \{\star, \diamond\}, \emptyset, \{2, \circ\}\}$

are systems of sets.

Subsets of a set

Definition (subset, proper subset)

A set A is called a **subset** of set B , in notation: $A \subseteq B$, if every element of A is also an element of B , that is if

$$\forall x (x \in A \Rightarrow x \in B)$$

If $A \subseteq B$ and $A \neq B$, then A is a **proper subset** of B , in notation: $A \subsetneq B$.

Note: Every set is a subset (but not a proper subset) of itself.

Proposition (Properties of the subset relation; proof is hw)

For every set A, B and C :

- 1 $\emptyset \subseteq A$.
- 2 $A \subseteq A$ (reflexivity).
- 3 $(A \subseteq B \wedge B \subseteq C) \Rightarrow A \subseteq C$ (transitivity).
- 4 $(A \subseteq B \wedge B \subseteq A) \Rightarrow A = B$ (anti-symmetry).

Defining sets using the set-builder notation

Subset axiom scheme (or Axiom schema of specification)

Let A be a set and $\mathcal{F}(x)$ be a formula (i.e. $\mathcal{F}$ is a 'property' that can be described by a formula). Then the collection of all those elements x of A for which $\mathcal{F}(x)$ is true (i.e. satisfy property $\mathcal{F}$) is a set. This set is denoted as

$$\{x \in A : \mathcal{F}(x)\} = \{x \in A \mid \mathcal{F}(x)\}.$$

Note:

- For $\{x \in A : \mathcal{F}(x)\}$ the notations $\{x : x \in A \wedge \mathcal{F}(x)\}$ and $\{x : x \in A, \mathcal{F}(x)\}$ are also commonly used.
- This notation (all the above variants) is often referred to as set-builder notation.

Examples:

- $\{x \in \mathbb{N} : 3 \leq x < 10\} = \{3, 4, 5, 6, 7, 8, 9\}.$
- $\{n \in \mathbb{Z} : \exists m (m \in \mathbb{Z} \wedge n = m^2)\}$: the set of square numbers.

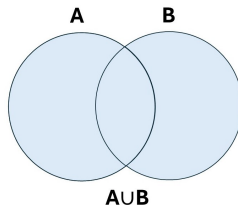
Set operations: set union

Definition (set union)

The **union** of two sets A and B , denoted by $A \cup B$ is the set consisting of those elements which belong to A or B (*at least one of them*), that is:

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

Venn diagram representation:



Examples:

- $\{a, b, c\} \cup \{b, c, d\} = \{a, b, c, d\}$
- $\{x \in \mathbb{R} : 0 < x\} \cup \{x \in \mathbb{R} : x < 0\} = \{x \in \mathbb{R} : x \neq 0\}$

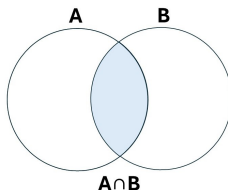
Set operations: set intersection

Definition (set intersection)

The **intersection** of two sets A and B , denoted by $A \cap B$, is the set containing exactly those elements that are elements of *both* A *and* B :

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

Venn diagram representation:



Examples:

- $\{a, b, c\} \cap \{b, c, d\} = \{b, c\}$
- $\{x \in \mathbb{R} : 0 < x\} \cap \{x \in \mathbb{R} : x < 0\} = \emptyset$

Union and intersection of 'many' sets

Definition (union of a system of sets)

Let $\mathcal{A}$ be a system of sets (i.e. a set in which the elements are also sets).

The **union of system** $\mathcal{A}$ denoted by $\bigcup \mathcal{A} = \bigcup \{A : A \in \mathcal{A}\} = \bigcup_{A \in \mathcal{A}} A$ is the set of all those elements which belong to at least one set in $\mathcal{A}$, that is

$$\bigcup \mathcal{A} = \{x \mid \exists A \in \mathcal{A} : x \in A\}.$$

Definition (intersection of a system of sets)

The **intersection of system** $\mathcal{A}$ denoted by $\bigcap \mathcal{A} = \bigcap \{A : A \in \mathcal{A}\} = \bigcap_{A \in \mathcal{A}} A$ is defined as

$$\bigcap \mathcal{A} = \{x \mid \forall A \in \mathcal{A} : x \in A\}.$$

Note: In particular: $A \cup B = \bigcup \{A, B\}$ and $A \cap B = \bigcap \{A, B\}$.

Examples:

- $\bigcup \{\{1, 4, 5\}, \{1, 2, 8\}, \{1, 3\}\} = \{1, 2, 3, 4, 5, 8\}$ and $\bigcap \{\{1, 4, 5\}, \{1, 2, 8\}, \{1, 3\}\} = \{1\}$.
- Let $I_n = \{x \in \mathbb{R} : n \leq x \leq n+1\}$, $\forall n \in \mathbb{Z}$ and $\mathcal{I} = \{I_n : n \in \mathbb{Z}\}$. Then $\bigcup \mathcal{I} = \mathbb{R}$ and $\bigcap \mathcal{I} = \emptyset$.

Disjoint and pairwise disjoint systems of sets

Definition ((pairwise) disjoint system of sets)

Two sets A and B are said to be **disjoint** if $A \cap B = \emptyset$.

In general: Let $\mathcal{A}$ be a system of sets.

- System $\mathcal{A}$ is said to be **disjoint** if $\bigcap \mathcal{A} = \emptyset$. (In this case we can also say that the elements of $\mathcal{A}$ are **disjoint**.)
- System $\mathcal{A}$ is said to be **pairwise disjoint** if $\forall A, B \in \mathcal{A}, A \neq B : A \cap B = \emptyset$. (Or we can also say that the elements of $\mathcal{A}$ are **pairwise disjoint**.)

System is pairwise disjoint. $\xrightarrow{\quad}$ System is disjoint.

Examples:

- The sets $\{1, 2\}$ and $\{3, 4\}$ are disjoint.
- The sets $\{1, 2\}$, $\{2, 3\}$ and $\{1, 3\}$ are disjoint, but **not** pairwise disjoint.
- The sets $\{1, 2\}$, $\{3, 4\}$ $\{5, 6\}$ are pairwise disjoint.
- Let $I_n = \{x \in \mathbb{R} : n \leq x \leq n+1\}$, $\forall n \in \mathbb{Z}$ and $\mathcal{I} = \{I_n : n \in \mathbb{Z}\}$. Then the system $\mathcal{I}$ is disjoint, but **not** pairwise disjoint.

Properties of set union and intersection

Properties of set union and intersection

For every set A , B and C :

- | | | |
|--------------------------------|--|--|
| 1. idempotence: | $A \cup A = A$ | $A \cap A = A$ |
| 2. associativity: | $A \cup (B \cup C) = (A \cup B) \cup C$ | $A \cap (B \cap C) = (A \cap B) \cap C$ |
| 3. commutativity: | $A \cup B = B \cup A$ | $A \cap B = B \cap A$ |
| 4. properties of $\emptyset$: | $A \cup \emptyset = A$ | $A \cap \emptyset = \emptyset$ |
| 5.: | $A \subseteq B \Leftrightarrow A \cup B = B$ | $A \subseteq B \Leftrightarrow A \cap B = A$ |

Proof

Properties of set union:

$$\textcircled{1} \quad x \in A \cup A \stackrel{(1)}{\Leftrightarrow} x \in A \vee x \in A \stackrel{(2)}{\Leftrightarrow} x \in A.$$

(1) definition of $\cup$

(2) idempotence of $\vee$

$$\textcircled{2} \quad x \in A \cup (B \cup C) \stackrel{(1)}{\Leftrightarrow} x \in A \vee x \in B \cup C \stackrel{(1)}{\Leftrightarrow} x \in A \vee (x \in B \vee x \in C) \stackrel{(2)}{\Leftrightarrow} (x \in A \vee x \in B) \vee x \in C \stackrel{(1)}{\Leftrightarrow} x \in A \cup B \vee x \in C \stackrel{(1)}{\Leftrightarrow} x \in (A \cup B) \cup C$$

(1) definition of $\cup$

(2) associativity of $\vee$

Properties of set union and intersection

Properties of set union and intersection

For every set A , B and C :

- | | | |
|--------------------------------|--|--|
| 1. idempotence: | $A \cup A = A$ | $A \cap A = A$ |
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| 3. commutativity: | $A \cup B = B \cup A$ | $A \cap B = B \cap A$ |
| 4. properties of $\emptyset$: | $A \cup \emptyset = A$ | $A \cap \emptyset = \emptyset$ |
| 5.:: | $A \subseteq B \Leftrightarrow A \cup B = B$ | $A \subseteq B \Leftrightarrow A \cap B = A$ |

Proof

Properties of set union:

- ③ *Homework: using commutativity of $\vee$, similar.*
- ④ $x \in A \cup \emptyset \stackrel{1}{\Leftrightarrow} x \in A \vee x \in \emptyset \stackrel{2}{\Leftrightarrow} x \in A \vee F \stackrel{3}{\Leftrightarrow} x \in A.$
 - (1) definition of $\cup$
 - (2) definition of $\emptyset$
 - (3) definition of $\vee$
- ⑤ $\Rightarrow: A \subseteq B \Rightarrow A \cup B \subseteq B$, but $B \subseteq A \cup B$ is always true, hence $A \cup B = B.$
 $\Leftarrow: \text{If } A \cup B = B$, then all elements of A are also elements of $B.$

Proof of the properties of set intersection: Similar, homework.

Distributivity of set union and set intersection

Proposition (Distributivity of set union and set intersection)

For every set A , B and C :

- ① $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (distributivity of $\cap$ over $\cup$)
- ② $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (distributivity of $\cup$ over $\cap$)

Proof

$$\begin{aligned} \textcircled{1} \quad x \in A \cap (B \cup C) &\stackrel{(1)}{\Leftrightarrow} x \in A \wedge x \in B \cup C \stackrel{(2)}{\Leftrightarrow} \\ &\stackrel{(2)}{\Leftrightarrow} x \in A \wedge (x \in B \vee x \in C) \stackrel{(3)}{\Leftrightarrow} (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \\ &\stackrel{(1)}{\Leftrightarrow} x \in A \cap B \vee x \in A \cap C \stackrel{(2)}{\Leftrightarrow} x \in (A \cap B) \cup (A \cap C) \end{aligned}$$

(1) definition of $\cap$

(2) definition of $\cup$

(3) distributivity of $\wedge$ over $\vee$

- ② HW. (Similar to 1.)

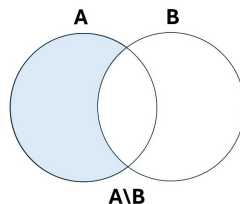
Set difference

Definition (set difference)

The **difference** of two sets A and B is denoted by $A \setminus B$ and is defined as

$$A \setminus B = \{x \in A : x \notin B\}.$$

Venn diagram representation:



Examples:

- $\{1, 2, 4, 8, 16\} \setminus \{1, 2, 3, 4, 5\} = \{8, 16\}$
- $\{1, 2, 3, 4, 5\} \setminus \{1, 2, 4, 8, 16\} = \{3, 5\}$
- $\{1, 2, 4, 8, 16\} \setminus \emptyset = \{1, 2, 4, 8, 16\}$
- $\mathbb{Z} \setminus \{2n \mid n \in \mathbb{Z}\} = \{2n + 1 \mid n \in \mathbb{Z}\}$

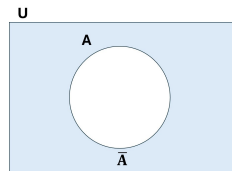
Set complement

Definition (complement of a set)

For a given universal set U and $A \subseteq U$, the **complement** of A is denoted by $\overline{A} = A'$ and is defined as

$$\overline{A} = A' = U \setminus A.$$

Venn diagram representation:



Proposition (Expressing set difference using set complement)

For every set A and B : $A \setminus B = A \cap \overline{B}$.

Proof

$$x \in A \setminus B \Leftrightarrow x \in A \wedge x \notin B \Leftrightarrow x \in A \wedge x \in \overline{B} \Leftrightarrow x \in A \cap \overline{B}$$

Properties of set complement

Proposition (Properties of set complement; proof hw)

Denote by U the universal set. Then for every set $A, B \subseteq U$ the following hold:

- 1 $\overline{\overline{A}} = A;$
- 2 $\overline{\emptyset} = U;$
- 3 $\overline{U} = \emptyset;$
- 4 $A \cap \overline{A} = \emptyset;$
- 5 $A \cup \overline{A} = U;$
- 6 $A \subseteq B \Leftrightarrow \overline{B} \subseteq \overline{A};$
- 7 $\overline{A \cap B} = \overline{A} \cup \overline{B};$
- 8 $\overline{A \cup B} = \overline{A} \cap \overline{B}.$

Properties 7 and 8 are called **De Morgan's laws**.

Proof ($\overline{A \cap B} = \overline{A} \cup \overline{B}$)

- 1 $x \in \overline{A \cap B} \Leftrightarrow \neg x \in A \cap B \Leftrightarrow \neg(x \in A \wedge x \in B) \Leftrightarrow$
 $\Leftrightarrow \neg(x \in A) \vee \neg(x \in B) \Leftrightarrow x \in \overline{A} \vee x \in \overline{B} \Leftrightarrow x \in \overline{A} \cup \overline{B}$

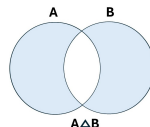
Symmetric difference of sets

Definition (symmetric difference)

The **symmetric difference** of two sets A and B is denoted by $A \triangle B$ and is defined as

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

Venn diagram representation:



Proposition (Alternative expression of the symmetric difference; proof HW)

For every set A and B : $A \triangle B = (A \cup B) \setminus (B \cap A)$.

Power set of a set

Definition (power set)

Let A be a set. The set of all subsets of A is called the **power set of A** and it is denoted by 2^A or by $\mathcal{P}(A)$.

- $A = \emptyset, 2^\emptyset = \{\emptyset\}$
- $A = \{a\}, 2^{\{a\}} = \{\emptyset, \{a\}\}$
- $A = \{a, b\}, 2^{\{a,b\}} = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Notation: For a finite set A , the number of elements in A is denoted by $|A|$.

Proposition (Size of the power set of a finite set; proof later)

For every finite set A we have $|\mathcal{P}(A)| = 2^{|A|}$.

Appendix: ZF and ZFC

A famous paradox in naive set theory

Why do we need axioms?

Russell's paradox (Bertrand Russell, 1872 – 1970)

Call every set that is not an element of itself **good**.

Call every set that is an element of itself **bad**.

Define set **A** as the set of all **good** sets.

Is **A** a **good** or a **bad** set?



- **A** is a **good** set. $\xRightarrow{\text{def. of } A}$ **A** is an element of itself. $\Rightarrow$ **A** is a **bad** set. ⚡
- **A** is **bad** set. $\xRightarrow{\text{def. of } A}$ **A** is not an element of itself. $\Rightarrow$ **A** is a **good** set. ⚡

The possible ways in which sets can be defined need to be restricted and clearly determined. $\Rightarrow$ **Axiomatic set theory**: Zermelo-Fraenkel set theory axioms.

The **axioms** of set theory define some basic properties of sets which do not need to be proved, but are accepted as true.

Appendix: Zermelo-Fraenkel axioms of set theory

Note: The Appendix contains the Zermelo-Fraenkel system of axioms and the Axiom of choice. This is not part of the compulsory syllabus, and hence, you will not be assessed on it. It is additional material which you can read optionally if interested.

Axiomatic set theory: Modern set theory is called axiomatic set theory, because it is derived from a set of so called *axioms*, i.e. statements which are accepted as true without proof. The theory is the collection of all theorems that can be inferred from the axioms (using the rules laid out in mathematical logic). There are different axiomatisations of set theory; the most widely used one is the Zermelo-Fraenkel (ZF) set of axioms, often complemented by the Axiom of choice, yielding ZFC.

Predicates in ZF (and ZFC)

The theory has two *predicates*, i.e. undefined basic concepts: that of a *set* and the relation to be an *element of* a set ($\in$).

The axioms describe certain properties of sets, many of them providing methods by which new sets can be constructed based on existing ones.

Zermelo-Fraenkel axioms of set theory

- ① **Axiom of extensionality:** Two sets are equal (are the same set) if they have the same elements:

$$\forall x \forall y (\forall z (z \in x \Leftrightarrow z \in y) \Rightarrow x = y)$$

- ② **Axiom of regularity (or axiom of foundation):** Every nonempty set contains an element that is disjoint from it:

$$\forall x (x \neq \emptyset \Rightarrow \exists y (y \in x \wedge y \cap x = \emptyset))$$

- ③ **Axiom schema of specification (Axiom schema of restricted comprehension):** For every set z and formula φ there exists a set y such that for every set x : x is an element of y if and only if x is an element of z and φ holds for x :

$$\forall z \forall w_1 \forall w_2 \dots \forall w_n \exists y \forall x (x \in y \Leftrightarrow ((x \in z) \wedge \varphi(x, w_1, w_2, \dots, w_n, z)))$$

(It is an axiom *schema*, since there is one axiom for each formula φ .)

Zermelo-Fraenkel axioms of set theory continued...

- ① **Axiom of pairing:** If x and y are sets, then there exists a set which contains both x and y as elements:

$$\forall x \forall y \exists z ((x \in z) \wedge (y \in z))$$

- ② **Axiom of union:** For every set of sets $\mathcal{F}$ there is a set A containing every element that is an element of some element of $\mathcal{F}$:

$$\forall \mathcal{F} \exists A \forall Y \forall x ((x \in Y \wedge Y \in \mathcal{F}) \Rightarrow x \in A)$$

- ③ **Axiom schema of replacement:** The image of a set under every definable function is also a set:

$$\forall w_1, \dots, w_n \forall A ((\forall x \in A \exists! y \varphi(x, y, w_1, \dots, w_n, A)) \Rightarrow \\ \exists B \forall y (y \in B \Leftrightarrow \exists x \in A \varphi(x, y, w_1, \dots, w_n, A)))$$

Zermelo-Fraenkel axioms of set theory continued...

- ① **Axiom of infinity:** For every set w abbreviate $w \cup \{w\}$, by $S(w)$. (Applying the Axiom of Pairing with $x = y = w$ so that the set z is $\{w\}$ we can see that $\{w\}$ is a set.) Then there exists a set X such that the empty set $\emptyset$ is an element of X and, whenever a set y is an element of X , $S(y)$ is also an element X :

$$\exists X (\exists e (\forall z \neg (z \in e) \wedge e \in X) \wedge \forall y (y \in X \Rightarrow S(y) \in X))$$

- ② **Axiom of the power set:** For every set x , there is a set y that contains every subset of x :

$$\forall x \exists y \forall z (z \subseteq x \Rightarrow z \in y)$$

The above constitute the Zermelo-Fraenkel system of axioms (ZF).

Axiom of choice and ZFC

Adding the Axiom of choice (AC) to ZF we obtain the system of axioms denoted by ZFC:

- **Axiom of choice:** For every set X of nonempty sets there exists a 'choice' function f which maps each set in X to an element of that set:

$$\forall X (\emptyset \notin X \implies \exists f: X \rightarrow \cup X \quad \forall A \in X (f(A) \in A)).$$